Big Bang and Creation were the same thing. *Fiat lux* would have to be changed to ‘Switch the light back on!’

To contest both Einstein and the Pope, convincing both that they were mistaken, and to be right in both cases, is surely something of a result. ‘The master’ lived up to his name.

Today confirmations are overwhelming: the universe, in a far-distant past, was extremely hot and extremely compact, and has expanded since. We can reconstruct in detail the history of the universe, starting with its initial hot, compressed state. We know how atoms, elements, galaxies and stars formed and how the universe as we see it today developed. Recent extended observations of the radiation that fills the universe carried out mainly by the Planck satellite once again confirmed in full the theory of the Big Bang. We know with a reasonable degree of certainty what happened on a large scale to our universe in the last 14 billion years, from the time when it was a ball of fire.

And to think that, initially, the phrase ‘theory of the Big Bang’ was coined by opponents of the theory, to mock an idea that seemed outlandish … Instead, in the end, we were all persuaded: 14 billion years ago the universe was a compressed ball of fire.

But what happened *before* this initial hot and compressed state?

Regressing in time, temperature increases, as does the density of matter and energy. There is a point at which they reach the Planck scale: 14 billion years ago. At that point, the equations of general relativity are no longer valid, because it is no longer possible to ignore quantum mechanics. We enter into the realm of quantum gravity.

Quantum cosmology

To understand what happened 14 billion years ago, therefore, quantum gravity is required. What do the loops tell us about the subject?

Consider an analogous but simpler situation. According to classical mechanics, an electron falling straight into an atomic nucleus would be swallowed by the nucleus and disappear. But this is not what happens in reality. Classic mechanics is incomplete, and it is necessary to take quantum effects into account. A real electron is a quantum object and does not follow a precise trajectory: it isn’t possible to keep it inside too small a region. The more it is concentrated, the more it slips rapidly away. If we want to stop it around the nucleus, the most we can do is to force it into an orbit of the size of the smallest atomic orbital: it could not stay any closer to the nucleus. Quantum mechanics prevents a real electron from falling into a nucleus. A quantum repulsion pushes away the electron when it gets too close to the centre. Thus, thanks to quantum mechanics, matter is stable. Without it,